

Los Angeles Regional Water Quality Control Board

June 20, 2022

Mr. Nathan Jacobsen
311 Main Road
Suite 1 N45V
Point Mugu, CA 93042
nathan.j.jacobsen.civ@us.navy.mil

GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS FOR THE SAN NICOLAS ISLAND WWTP AT BUILDING 27, SAN NICOLAS ISLAND, CALIFORNIA 93042 (FILE NO. 20-065, WDR ORDER WQ 2014-0153-DWQ, SERIES NO. 057, CI NO. 10667, GLOBAL ID WDR100052324)

Dear Mr. Jacobsen:

The Los Angeles Regional Water Quality Control Board (Regional Water Board) is the public agency with primary responsibility for the protection of ground and surface water quality for all beneficial uses of water within major portions of Los Angeles and Ventura Counties. The subject property is within the Regional Water Board jurisdiction.

Summary of the Technical Reports

The Department of the Navy, U. S. Naval Air Weapons Station, Point Mugu is authorized to discharge from the San Nicolas Island (SNI) Wastewater Treatment Plant (WWTP) under individual waste discharge requirements (WDRs) Order No. 94-006 issued by the Regional Water Board in 1994. The WWTP serves a small Naval facility on the San Nicolas Island with an average population of 240 Navy personnel and contractors, the majority of which are temporary or transient residents.

The WWTP consists of three recirculated ponds with a chlorination system that discharges to two nearby land application areas (LAAs), as shown in Figure 1. The design capacity of the WWTP is 40,000-gallons per day (gpd) but typically does not exceed 15,000 gpd.

On July 1, 2020, the Regional Water Board received your application, including a technical report, for enrollment under *General Order No. WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems (General WDRs)*, adopted by the State Water Resources Control Board on September 23, 2014. The application identified the Naval Base Ventura County as the owner of the WWTP and the Naval Facilities Engineering Command Southwest as the operator.

In October 2021, the Regional Water Board received your *Technical Report Addendum 1 - Additional Change in Design and Operation Information* in response to Regional Water Board's comment on required mitigation measures for handling a 100-year storm event. Addendum 1 provides the required information outlined in the General WDRs to justify land application of wastewater that is not disinfected prior to land application, and includes an overview of SNI WWTP designs, operations, and operational changes in support of enrolling the SNI WWTP under the General WDRs Permit.

The WWTP is currently classified by the State Water Resources Control Board (State Water Board) as a Class II System due to chlorination (disinfection) of treated wastewater prior to land disposal (Figure 2). The treatment and operational changes described in this application illustrate the system changes will meet the General WDRs' LAA containment requirements, thereby enabling the SNI WWTP to move from Class II to Class I per the State Water Board WWTP classification system. You are requesting to decommission the chlorination system and upgrade the LAAs in order for the WWTP to be reclassified as a Class I system. Figure 3 shows a summary of changes to the current treatment process that support this reclassification.

Due to the oversized nature of the WWTP relative to the influent volume and the arid climate of San Nicolas Island, most water removal from the ponds occurs via evaporation. When discharge occurs, it is only needed to control freeboard of the ponds during the springtime rainy season, and discharge to LAAs (Figure 1) has never been needed for some years. The LAAs comprise 1.3 acres of cleared land surrounded by 9-inch-tall earthen berms (Figure 4). You also propose to install French drains at the low side of each LAA which drain back into the treatment ponds (Figure 5). These changes are proposed to meet requirements to contain stormwater runoff from LAAs where undisinfected effluent is applied.

Summary of Regional Water Board Review and Requirements

The Regional Water Board staff has reviewed the information provided and has determined that the Site can be enrolled under the *General Order No. WQ 2014-0153-DWQ, General Waste Discharge Requirements (WDRs) for Small Domestic Wastewater Treatment Systems*, adopted by the State Water Resources Control Board on September 23, 2014.

Enclosed are your General WDRs, consisting of Order WQ 2014-0153-DWQ (Order), Monitoring and Reporting Program (MRP) No. CI-10667, and Standard Provisions Applicable to WDRs (Attachment 2). Based on the location of the Site and anticipated performance of the proposed changes to the WWTP, the requirements specified in Order WQ 2014-0153-DWQ, including prohibitions specified in Section A and requirements for all wastewater systems specified in Section B.1, specific requirements for pond systems in Section B.5, land application requirements in Section B.7 and Sludge/Solids Disposal requirements in Section B.8, are applicable to your discharge. Groundwater and surface water limitations outlined in Section C and effluent limitations in Section D are also applicable. Additionally, the effluent shall meet the following limitations equivalent to the groundwater quality objectives specified in the Basin Plan for San Nicolas Island

groundwater basin: 1100 mg/l for total dissolved solids, 150 mg/l for sulfate, and 350 mg/l for chloride. The maximum allowable effluent volume shall be 20,000 gpd. At this time, groundwater monitoring is not required.

The Site discharges to the San Nicolas Island groundwater basin (Basin) through two LAAs. The Basin is not classified as a nitrogen degraded basin and since discharge from the WWTP is limited to less than 20,000 gpd, nitrogen effluent limits are not required by the General WDRs at this time. The WWTP has a chlorination system, but the Navy proposes to upgrade the LAAs and cease chlorination to prevent chemical waste due to the infrequent discharge from the facility. Until the proposed upgrades to the LAAs, illustrated in Figure 5, are completed, effluent must be chlorinated, and chlorine residual monitoring specified in MRP No. CI-10667 is required when discharging the wastewater to LAAs. Once the LAA upgrades have been completed, chlorination is no longer required, and monitoring of chlorine residual is not required so long as no chlorination of effluent occurs. You are required to include progress of the proposed LAA upgrades as part of your monthly monitoring reports.

Should changes affecting the operation of the WWTP at the Site be needed, revised engineering drawings showing the changes must be filed with the Regional Water Board a minimum of thirty days prior to the change. You must receive an approval from the Regional Water Board prior to making any changes to the Site.

The MRP requires you to implement the monitoring program on the effective date of coverage under this permit. When submitting monitoring or technical reports to the Regional Water Board per these requirements, please include a reference to "Compliance File No. CI-10667", which will assure that the reports are directed to the appropriate file and staff. Also, please do not combine other reports with your monitoring reports. Submit each type of report as a separate document.

You shall comply with the Electronic Submittal of Information (ESI) requirements by submitting all reports required under the MRP, including monthly water usage, and pdf format monitoring reports to the State Water Resources Control Board GeoTracker database under Global ID WDR100052324.

Please see Paperless Office Notice for GeoTracker Users, dated December 21, 2011, for further details at:

<http://www.waterboards.ca.gov/losangeles/resources/Paperless/Paperless%20Office%20for%20GT%20Users.pdf>

To avoid paying future annual fees, please submit a written request for termination of your enrollment under the General WDRs in a separate letter if the permit is no longer needed. Be aware that the annual fee covers the fiscal year billing period beginning July 1 and ending June 30, the following year. You will pay the full annual fee if your request for termination is made after the beginning of the new fiscal year beginning July 1.

If you have any additional questions, please contact the Project Manager, Mr. Ryan Nickerson at (213) 620-6119 (ryan.nickerson@waterboards.ca.gov) or the Groundwater

Mr. Nathan Jacobsen
San Nicolas Island Wastewater Treatment Plant

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Permitting Unit Supervisor, Dr. James Kang, P.E. at (213) 576-6683
(jim.kang@waterboards.ca.gov).

Sincerely,

Renee Purdy
Executive Officer

Enclosures:

General Waste Discharge Requirements [Order No. WQ 2014-0153-DWQ](#) with the Standard Provisions Applicable to WDRs are available at the Regional Water Board website:

https://www.waterboards.ca.gov/losangeles/board_decisions/adopted_orders/index.html#2

Monitoring and Reporting Program No. CI-10667

Figure 1 – San Nicolas Island Wastewater Treatment Plant and Land Application Areas

Figure 2 – Existing Treatment system process flow diagram

Figure 3 – Treatment system process flow diagram with proposed changes

Figure 4 – Photograph of Land Application Area

Figure 5 – Design of proposed french drains

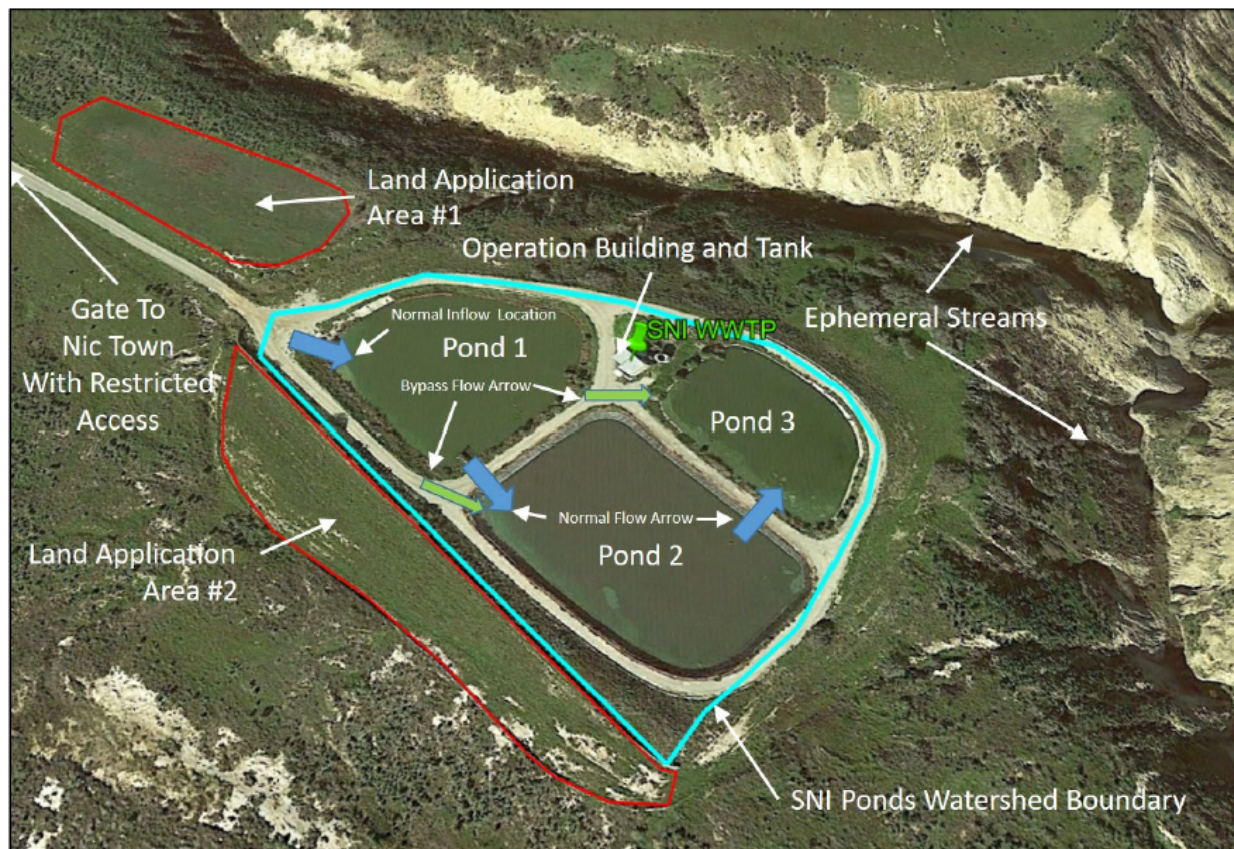


Figure 1. San Nicolas Island Wastewater Treatment Plant and Land Application Areas

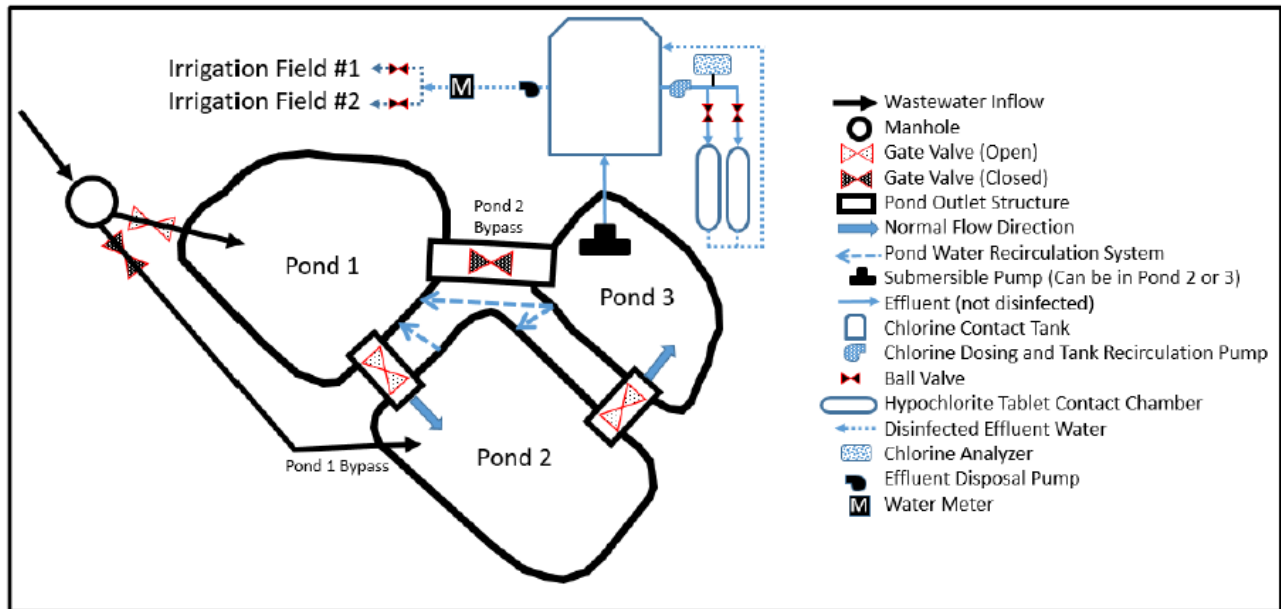


Figure 2. Existing SNI WWTP Process Flow Diagram

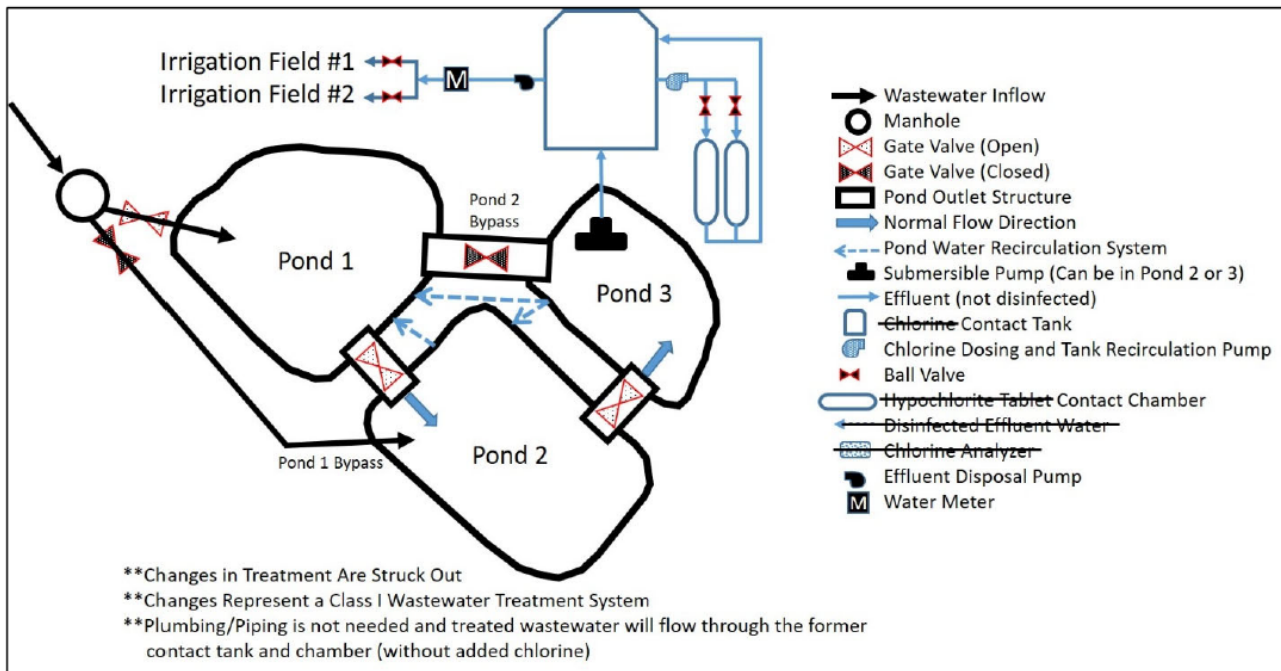


Figure 3. SNI WWTP Process Flow Diagram with proposed changes to existing wastewater treatment process



Figure 4. Photograph of Land Application Area #1 showing the edge with earthen berm to contain runoff, taken during September 28, 2021 inspection. Proposed changes shown in Figure 5 were not completed at the time of this inspection.

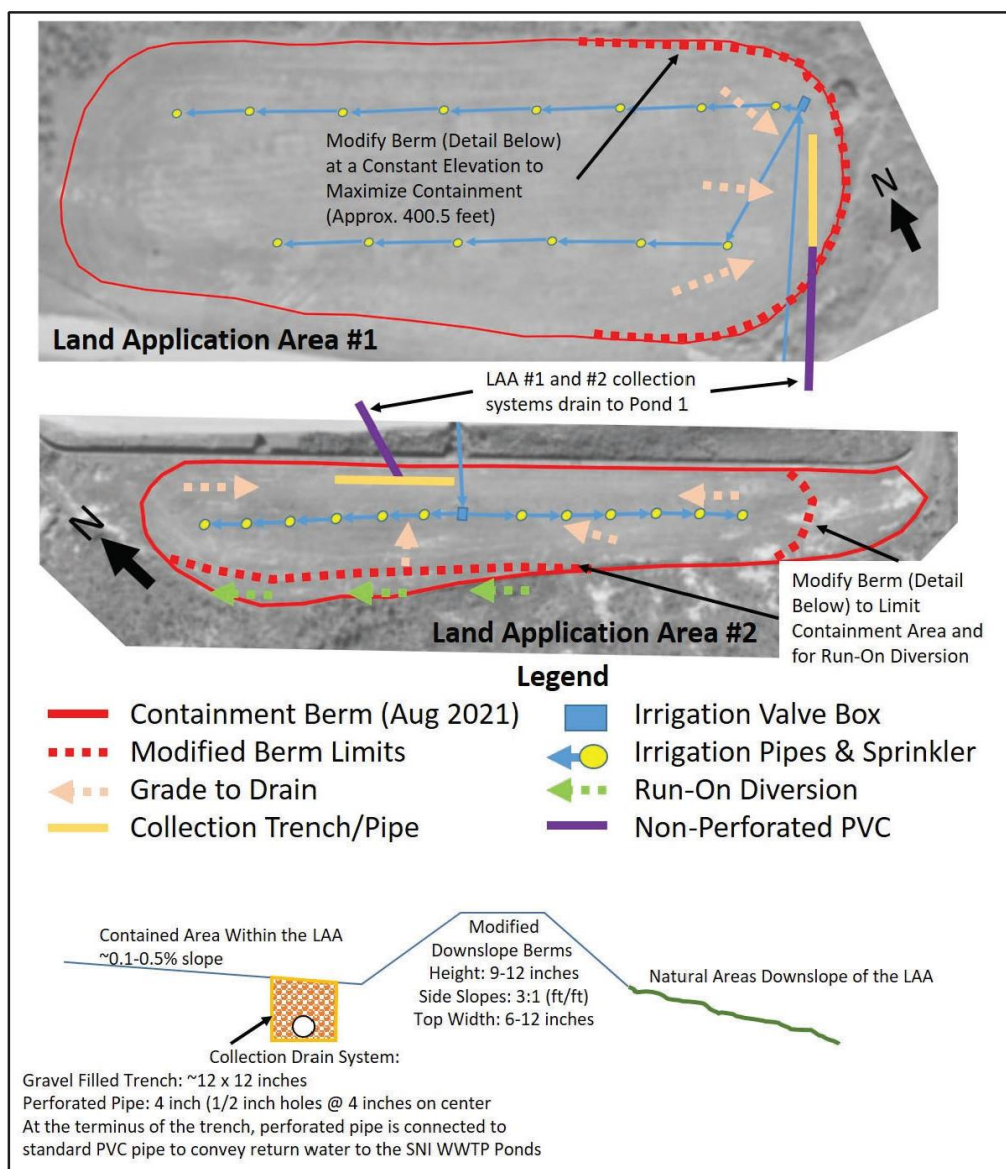


Figure 6 – LAA Berm Modification Designs and Collection and Return System

Figure 5: Map of LAAs and proposed modifications to berms and collection drain systems